2. Field Service Report (FSR) - Data Attribute Extraction

User Story

As a Data Scientist, I want to conduct an experiment to perform OCR on any scanned PDFs and extract metadata of FSRs to standard tables. I want to confirm what data enrichment should be done to get FSR data to AI ready stat such that the system can accurately retrieve specific pages & tables.

Acceptance Criteria

1. OCR Implementation
2. Metadata Extraction
3. Confirmation of data enrichment needs
4. Confluence Documentation Sign Off

All documentation of Data Attribute Extraction & FSR Scraping is located here.

FSR Scraping & Metadata Extraction – Proof of Concept Documentation

1. Introduction

This Proof of Concept (POC) was initiated to automate the extraction, normalization, and enrichment of metadata from Field Service Reports (FSRs). The goal is to streamline the process of converting unstructured PDF documents into structured datasets that can be used for analytics, reporting, and downstream integration with enterprise systems.

The effort was divided into three major stages:

1. **Extraction** – Parsing PDF files to capture key fields and metadata.
2. **Normalization** – Using AI/LLM models to standardize the extracted data into a consistent schema.
3. **Enrichment** – Integrating normalized data with enterprise reference tables (IBAT Equipment Master and Event Vision SOT) to fill gaps and enhance quality.

2. Stage 1 – PDF Extraction

Objective

Extract raw metadata from FSR PDF files in a flexible manner, capturing both titles and key-value fields.

Implementation

- **Tooling:** Implemented in Python using the pdfplumber library.
- **Process:**
  - Iterates through all PDFs in a designated folder.
  - Extracts text from the first page of each PDF.
  - Captures the **title** (first 2–3 lines before any colon-separated fields).
  - Identifies and stores **key:value pairs**, supporting multi-line values.
  - Handles duplicate keys by concatenating values.
- **Output:** JSON file containing a list of extracted records, each with:
  - PDF identifier (path/name)
  - Title
  - Key-value metadata fields

Key Features

- Robust error handling (logs errors but continues processing).
- Configurable limits (process all PDFs or restrict to a subset).
- Flexible parsing logic to handle variations in PDF formatting.

3. Stage 2 – Data Normalization

Objective

Transform the raw extracted metadata into a structured, standardized schema suitable for analysis, with refined handling of EV/EVP/EV Equipment Event IDs.

Implementation

- **Tooling:** Python + litellm library for LLM integration.
- **Process:**
  - Loads extracted JSON records.
  - Processes records in batches (default: 50 PDFs per batch).
  - Sends batch data to an LLM with a detailed prompt specifying:
    - Required fields (e.g., ESN, Equipment Sys ID, Event Type, Report Dates).
    - Mapping rules for project IDs:
      - Oracle Project IDs OFS Event ID (excluding EV/EVP prefixes).

        - EV- EV Equipment Event ID.
        - EVP- EV Project ID.
        - FSP- FSP Project ID.
        - XXX prefixes (A-, C-, etc.) xxx project id.
      - Controlled vocabulary for Event Type.
      - Splitting logic for multiple ESN/Equipment Sys IDs.
    - Receives normalized JSON output from the LLM.
    - Converts results into both JSON and Excel formats.

Output

- Normalized JSON with consistent schema.
- Excel file for easy review and downstream processing.

Key Features

- Intelligent handling of ambiguous fields (infers values from context).
- Strict enforcement of allowed values for Event Type.
- Splitting of complex records into multiple rows while preserving data integrity.
- Updated mapping logic for EV/EVP/EV Equipment Event IDs.

4. Stage 3 – Data Enrichment

Objective

Enhance normalized FSR metadata by joining with enterprise reference data (IBAT Equipment Master and Event Vision SOT).

Implementation

- **Tooling: Python + PySpark.**
- **Process:**
    - Loads normalized Excel into Pandas, then converts to Spark DataFrame.
    - Normalizes column names (ESN, Equipment Sys ID, etc.).
    - Cleans and strips prefixes (EV-, EVP-, FSP-) for join keys.
    - Joins FSR dataset with IBAT Equipment Master table on equipment_sys_id or ESN.
    - Fills missing values for ESN, Equipment Type, and Equipment Sub-Class using IBAT data.
    - Joins enriched dataset with Event Vision SOT table using EV, EVP, OFS, and FSP keys.
    - Fills missing values for project IDs, event types, and outage dates from Event Vision SOT.
    - Restores original column order for consistency.
    - Exports enriched dataset as a plain text file (CSV-style).

Output

- Final enriched dataset in text format, ready for downstream analytics.

Key Features

- Ensures completeness by filling gaps from IBAT and Event Vision SOT reference data.
- Maintains row order and original schema for traceability.
- Enhanced join logic with multiple keys (EV, EVP, OFS, FSP).
- Prefix restoration for IDs when filling missing values.

5. Results

The enriched data obtained after extracting the data attributes from FSR reports, normalizing the data across FSR reports and enriching the same with IBAT and Event Vision SOTs is provided in the csv format. The output is attached to this documentation and as well can be accessed from the following link: fsr_enriched_2016_Vinayaka_230.csv.

Currently, the FSR # column is largely unpopulated because most FSR reports do not include this information. During the earlier discussions with APM (Mohan Kumar), there was no indication of an alternate source (such as IBAT or Event Vision SOT) from which we could enrich this field, unlike other columns where enrichment was possible. As a result, only the few rows that originally carried FSR # values remain populated, and the rest cannot be filled through the current enrichment process.

*As finalized after discussion with APM, we will **verify extraction of this field (FSR #) once all files are loaded**, especially for the newer population where the field should be present.*

6. Achievements

- Automated extraction of metadata from hundreds of FSR PDFs.
- Standardized schema creation using AI-assisted normalization.
- Integration with enterprise equipment master data for enrichment.
- Multi-format outputs (JSON, Excel, text) for flexibility.
- Modular pipeline design, allowing independent improvements at each stage.

7. Conclusions

The POC successfully demonstrates the feasibility of automating FSR metadata extraction and enrichment. By combining PDF parsing, AI-driven normalization, and Spark-based enrichment, the pipeline transforms unstructured reports into structured, high-quality datasets. This lays the foundation for improved analytics, reporting, and operational efficiency in handling FSR data.